

Kenneth Emata

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SKILLS

- **Languages:** C/C++, Rust, Java, Python, JavaScript, SQL, R, Bash
- **Tools & Platforms:** Git/GitHub, Docker, VS Code, Linux, MATLAB, NeoVim, MS Office, Google

WORK EXPERIENCE

Classroom Facilitator/Curriculum Developer

June 2025 - Present

STEMA Learning Centre, Surrey, BC

- Designed and delivered technical curriculum for 30+ students ages 4–13, translating programming and engineering fundamentals into structured, age-appropriate lessons.
- Built programming modules in Python, JavaScript, HTML/CSS, and Scratch covering core concepts including loops, conditional logic, and variables.
- Led hands-on engineering lessons using robotics kits, focusing on mechanical assembly, structural stability, and sensor integration

SOFTWARE PROJECTS

Cellar: CLI Password Manager

September 2026

Technologies: Python, Typer, Argon2id, AES-256-GCM, Git/GitHub

- Built a CLI password manager from scratch that encrypts the vault with AES-256-GCM and derives keys with Argon2id.
- Stores Argon2id cost parameters per vault, so KDF difficulty can be raised later without breaking older vaults.
- Supports master-password rotation by re-encrypting the full vault under a newly derived key.
- Prevents data loss from crashes or concurrent access using atomic writes and a file lock.

Connect Four: Multiplayer Networked Game

April 2026

Technologies: Python, TCP Sockets, Threading, Tkinter, JSON

- Built a two-player Connect Four game where a server accepts two clients on dedicated threads over raw TCP sockets and broadcasts board updates to both players.
- Kept game state, move validation, and win detection server-side, with a Tkinter client that renders the board from incoming JSON messages.

Turtle Trips: AI-Powered Travel Planner

October 2025

Technologies: Flask, Python, HTML/CSS, Javascript, Git/Github, Gemini API

- Built a Flask app that turns natural language prompts into multi-day itineraries covering dining, lodging, and activities using the Gemini API.
- Integrated Google Maps/Places APIs to plot waypoints, calculate route distances, and pull in real-time venue data.
- Led version control and code review for a 4-person team at SFU's StormHacks 2025.

MarketMimic: Stock Trading Simulator

July 2024

Technologies: Java, HTML/CSS, JavaScript, Maven, Websocket Docker

- Built a Java web app with a team of 5 where first-time investors place simulated trades with virtual currency against real-time Finnhub market data streamed over WebSockets.
- Implemented a watchlist of up to 50 tickers with interactive Chart.js price history

EDUCATION

Bachelor of Science in Computing Science

Expected: December 2027

Simon Fraser University, Burnaby, BC